COMPARABLE PROJECTS VALUATION ENGINE

Technical Specification, Column Logic & Mathematical Reference Guide

Platform: Real Estate Valuation POC | **Architecture:** Elastic Valuation Engine v2.4 | **Date:** October 2026
Scope: Complete Logic, Verification Rules, Formulas & Edge-Cases for all 25 Table and CSV Columns

Executive Summary & Underwriting Standard: The Comparable Projects Valuation Engine evaluates candidate residential projects through evidence-grounded spatial discovery, live portal web intelligence, and multi-dimensional elastic similarity scoring. It enforces a **Strict Zero-Imputation Policy** (missing public rates or areas are never fabricated or guessed) and institutional qualification rules that eliminate unverified entities.

1. Architectural Pipeline & Discovery Stages

Stage 1: Input & Concurrent Geocoding	Resolves user input (Project Name or Lat/Lng coordinates) via Google Places API in < 0.1s. Dispatches 3-way concurrent execution across Comparables, CBD Deep Agent, and Amenities.
Stage 2: Automated Baseline Calibration	Dispatches targeted intelligence search for the subject property to extract benchmark rate, carpet area, unit configuration, developer tier, and building age/possession date.
Stage 3: Spatial Discovery & Strict Asset Filtering	Scans 2.0 to 10.0 km radius. Employs asset bifurcation (Flat / Shop / Office / Others) and commercial tenant filtering to isolate genuine assets from corporate service companies.
Stage 4: Multi-Tier Search & Persistent Circuit Breaker	Executes high-concurrency search (8 threads) with persistent quota circuit breakers (.quota_status.json), automatically routing around exhausted keys with zero retry latency to DuckDuckGo.
Stage 5: Dual-Pass Project DNA & Age Extraction	Parses rates (including Commercial Income Capitalization @ 8% cap rate / 150x rent multiplier), carpet areas, unit typologies, and dedicated Building Age & Possession Year timelines.
Stage 6: Elastic Valuation & Market Banding	Calculates continuous Gaussian and exponential scores across 7 physical and financial dimensions with dynamic elastic re-weighting when price is unlisted.

2. Subject Property Baseline Calibration

2.1 How the Source Benchmark Rate is Derived (4-Tier Hierarchy)

The subject property rate (**Source Rate**, *R_src*) anchors the entire valuation comparison. The engine establishes this benchmark through an automated 4-tier decision cascade:

Priority 1: User Manual Override	If the appraiser supplies an explicit benchmark rate in the sidebar under 'Optional Manual Overrides', this rate takes absolute authority, overriding all automated web scraping. Indicated by '(Custom Override)'.
Priority 2: Direct Disclosed Web Rate	Extracted directly from top real estate portals (MagicBricks, 99acres, Housing.com, SquareYards). If a range is quoted (e.g. ■7,500 to ■9,200/sq.ft), the engine computes the exact midpoint: R_src = (Min + Max) / 2.
Priority 3: Ticket Size ÷ Carpet Area	When portals report total apartment ticket prices (e.g., ■75.61 L) and carpet areas (e.g., 540 sq.ft) but no per-sqft rate, the engine computes: R_src = round(Total INR Ticket / Carpet Area, 0).
Priority 4: Locality Cluster Median Fallback	If the subject site is unlisted or newly launched with no public web DNA, the engine queries all verified nearby residential projects discovered in the radius and applies their statistical cluster median: R_src = Median({R_comp}). Indicated by '(Locality Cluster Median)'.

2.2 How Source Price is Resolved When Coordinates (Lat / Long) are Input

When the user inputs GPS coordinates directly (e.g., Latitude 19.066000, Longitude 72.860000):
1. **Reverse-Geocoding:** The coordinates are passed to Google Maps Geocoding API to resolve the formatted street address, sublocality, and building name.
2. **Building Extraction:** If the coordinate pin falls on an established apartment society (e.g., *Kalpataru Sparkle*), that name is extracted as the subject entity.
3. **Targeted Search:** Portal queries are dispatched for that specific building to retrieve live rates.
4. **Unlisted / Vacant Site Handling:** If the coordinate represents an open plot or unlisted land parcel, the engine discovers verified residential societies within 2.0 km and computes the **Locality Cluster Median** as the baseline rate.

2.3 Allowable Price and Carpet Area Tolerance Corridors

**Once R_src and A_src are established, the engine sets strict institutional boundaries:
• Allowable Price Range (80%–120%): Min Rate = R_src × 0.80 | Max Rate = R_src × 1.20
  Example: For R_src = ■8,333/sq.ft ➔ Range = [■6,666, ■10,000 / sq.ft]
• Allowable Carpet Area Range (±15%): Min Area = A_src × 0.85 | Max Area = A_src × 1.15
  *Example: For A_src = 1,000 sq.ft ➔ Range = [850, 1,150 sq.ft]***

2.4 Target Property Configuration & Typology Filtering

Users can explicitly choose the target configuration: **Flat (1 BHK, 2 BHK, 3 BHK, 4 BHK, > 4 BHK / Penthouse)**, **Commercial Shop**, **Commercial Office**, or **Other / Mixed-Use**. When selected, the discovery engine enforces configuration-specific targeting:

Target Configuration	Property Type & Search Target	Strict Matching Evidence Rule
Flat - 1 BHK	Residential Flat	Matches 1 BHK, 1 Bed, Studio
Flat - 2 BHK	Residential Flat	Matches 2 BHK, 2 Bed
Flat - 3 BHK	Residential Flat	Matches 3 BHK, 3 Bed
Flat - 4 BHK	Residential Flat	Matches 4 BHK, 4 Bed
Flat - > 4 BHK	Residential Luxury	Matches 5 BHK, 6 BHK, Penthouse, Duplex, Villa
Shop (Commercial Retail)	Commercial Retail	Shopping malls, retail complexes, showrooms
Office (Commercial Corporate)	Commercial Office	IT tech parks, business centers, corporate offices
Auto-Detect	Dynamic from Web DNA	Discovered automatically from subject project

3. Comprehensive Column-by-Column Specification (All 25 Fields)

The directory table presents 22 core underwriting columns, and the downloadable CSV contains 25 comprehensive audit fields. Below is the exact data source, logic, formula, and display format for every single column:

Column 1: Name	**Data Source:** Google Places API (Nearby Search & Text Search) via clean_project_name()
Underwriting Logic & Rules:	Extracts entity name, strips commercial noise, marketing taglines ('Luxurious Apartments', 'Showroom for Sale'), city suffixes (', Pune', '- Mumbai'), and consolidates sub-units to parent society.
Mathematical Formula / Rule:	Regex sanitization: re.sub(r'\s*[-|–]\s*(?:Best|Premier|Luxurious|Top|Affordable|Offices?).*$', '', name)
Display Format / Output:	**Clean string (e.g., 'Gokulam by Sarvome', 'Vasant Valley')**
Column 2: Distance (km)	**Data Source:** Spherical coordinates calculation (Haversine Formula)
Underwriting Logic & Rules:	Computes great-circle distance between Subject Site coordinates (lat1, lng1) and Comparable Site coordinates (lat2, lng2).
Mathematical Formula / Rule:	$d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta\phi/2) + \cos(\phi1)\cdot\cos(\phi2)\cdot\sin^2(\Delta\lambda/2)}))$, with R = 6,371.0 km
Display Format / Output:	**Two-decimal float with unit (e.g., '0.79 km')**
Column 3: Verified Rate (■/sqft)	**Data Source:** Multi-tier live web search parsed via extract_project_dna_dynamic()
Underwriting Logic & Rules:	Strict Zero-Imputation Policy: Never fabricated or guessed. Evaluates prefix rates ('Price per sq. ft.: ■3,400 to ■4,100'), standard per-sqft strings, or ticket price ÷ carpet area. Bounds: [■2,500, ■150,000].
Mathematical Formula / Rule:	Direct rate, or (Min + Max)/2 if range, or round(Total INR Ticket / Carpet Area, 0)
Display Format / Output:	**'■ 4,039' or 'Rate Unlisted'**
Column 4: In 80%–120% Range?	**Data Source:** Comparison between Comparable Verified Rate (R_comp) and Subject Benchmark (R_src)
Underwriting Logic & Rules:	Computes percentage difference ΔP = (R_comp - R_src) / R_src. Categorizes into In-Range, Premium, Value, or Physical Match.
Mathematical Formula / Rule:	If 0.80·R_src <= R_comp <= 1.20·R_src ➔ In Range; If > 1.20 ➔ Premium (+ΔP%); If < 0.80 ➔ Value (-ΔP%)
Display Format / Output:	**'■ In 80%–120% Range', '■ Premium (+28.4%)', '■■ Value (-22.1%)', or '■ Physical Match (Rate Unlisted)'**
Column 5: Price Evidence	**Data Source:** Unaltered verbatim excerpt from the source real estate portal snippet
Underwriting Logic & Rules:	Extracts 120-character contextual snippet surrounding the discovered price token for full auditability.
Mathematical Formula / Rule:	Regex contextual window extraction from scraped portal text
Display Format / Output:	**Verbatim text: '...Price starts from ■75.61 L... (Total ■75.61 L / 1872 sq.ft)' or '—'**
Column 6: Carpet Area (sqft)	**Data Source:** Portal floor plans, brochure specifications, and RERA registration filings
Underwriting Logic & Rules:	Matches target unit configuration area (e.g. '2 BHK 534 to 544 sq.ft'), keyword 'carpet area', or floor plan area tags. Validated within [350, 3,500 sq.ft].
Mathematical Formula / Rule:	Area value in sq.ft, or midpoint (A1 + A2)/2 if range
Display Format / Output:	**Formatted integer (e.g., '1,872 sq.ft') or '—' if unlisted**

Column 7: Area Match	**Data Source:** Mathematical divergence comparison against Subject Carpet Area (A_src)
Underwriting Logic & Rules:	Computes area divergence ratio δA = |A_comp - A_src| / A_src against standard 15% tolerance threshold.
Mathematical Formula / Rule:	If δA <= 0.15 ➔ '■ In Range ({A_comp} sq.ft)'; Else ➔ 'Area ({A_comp} sq.ft)'; If unlisted ➔ 'Area Unlisted'
Display Format / Output:	**'■ In Range (950 sq.ft)', 'Area (1,450 sq.ft)', or 'Area Unlisted'**

Column 8: Carpet Area Evidence	**Data Source:** Verbatim source portal snippet declaring carpet area specifications
Underwriting Logic & Rules:	Captures the exact text context where unit dimensions or carpet area were published.
Mathematical Formula / Rule:	Contextual regex window from portal metadata
Display Format / Output:	**Verbatim text: '...Carpet Area: 534 to 544 sq. ft...' or '—'**

Column 9: Unit Typology	**Data Source:** Web snippet parser scanning layout configurations
Underwriting Logic & Rules:	Scans for 1 BHK, 2 BHK, 3 BHK, 4 BHK, 5 BHK, 6 BHK, Studio, Penthouse, Duplex, Villa, Shop, Office configurations.
Mathematical Formula / Rule:	Multi-typology token set extraction
Display Format / Output:	**Comma-separated list (e.g., '2 BHK, 3 BHK') or '—'**

Column 10: Unit Match	**Data Source:** is_unit_matching() comparing candidate available units against target configuration
Underwriting Logic & Rules:	Tests whether candidate explicitly discloses the target unit configuration. Mismatches disqualify the candidate and assign zero unit score.
Mathematical Formula / Rule:	If target unit matched ➔ '■ Matches ({unit})'; If candidate has other units ➔ '■■ Mismatch ({units})'; If unlisted ➔ 'Typology Unlisted'
Display Format / Output:	**'■ Matches (2 BHK)', '■■ Mismatch (3 BHK, 4 BHK)', or 'Typology Unlisted'**

Column 11: Unit Evidence	**Data Source:** Contextual portal text declaring unit configurations
Underwriting Logic & Rules:	Provides snippet proof for available room configurations in the project.
Mathematical Formula / Rule:	Contextual regex capture
Display Format / Output:	**Verbatim snippet: '...Configurations: 2 BHK, 3 BHK Apartments...' or '—'**

Column 12: Developer	**Data Source:** Entity title parsing and national/regional developer ontology
Underwriting Logic & Rules:	Scans parenthetical/prefix titles ('Project by Developer'), known brand dictionary, and structured markdown tags ('Developer: XYZ').
Mathematical Formula / Rule:	Ontology lookup + regex matching: r'\b(?:developer|builder|promoter)\s*[:*]*\s*([A-Za-z0-9\s&,\.]{3,50}?)'
Display Format / Output:	**Clean builder title (e.g., 'HRH City Projects Private Limited', 'Godrej Properties') or '—'**

Column 13: Developer Tier	**Data Source:** Institutional builder classification ontology
Underwriting Logic & Rules:	Classifies developer scale into Tier 1 (National Listed/Institutional), Tier 2 (Reputed Regional Multi-Project), or Tier 3 (Local/Standalone Society).
Mathematical Formula / Rule:	Lookup across KNOWN_BRAND_MAP registry
Display Format / Output:	**'Tier 1 (National Listed)', 'Tier 2 (Reputed Regional)', 'Tier 3 (Local / Standalone)', or '—'**

Column 14: Developer Evidence	**Data Source:** Source declaration confirming builder identity
Underwriting Logic & Rules:	Indicates whether developer was extracted from title, ontology registry, or portal snippet.
Mathematical Formula / Rule:	Citation path string
Display Format / Output:	**'Extracted from project name: by Sarvome' or verbatim snippet**

Column 15: Construction Status	**Data Source:** Portal possession timelines and RERA status declarations
Underwriting Logic & Rules:	Distinguishes Ready to Move (RTM) from Under Construction (UC) with direct evidence extraction.
Mathematical Formula / Rule:	If UC / Ongoing / Launching → 'Under Construction'; If RTM / Completed / Handover → 'Ready to Move'
Display Format / Output:	**'Ready to Move', 'Under Construction', or 'Status Unlisted'**

Column 16: Status Evidence	**Data Source:** Source snippet quoting possession, delivery, or construction milestone
Underwriting Logic & Rules:	Verbatim proof of project construction progress and operational delivery status.
Mathematical Formula / Rule:	Contextual regex extraction
Display Format / Output:	**Verbatim snippet: '...Ready to Move (Completed: 2023)...' or '—'**

Column 17: Building Age	**Data Source:** Completion date, year built, and property age declarations
Underwriting Logic & Rules:	Calculates building age from completion/possession year (Age = Current Year - Year Built). For under-construction assets, displays scheduled delivery timeline.
Mathematical Formula / Rule:	If UC → 'Under Construction (Possession: YYYY)'; If RTM → '{Current Year - Poss Year} Years'; If recent → '< 1 Year (Recent Possession)'
Display Format / Output:	**'3 Years', '5 Years', 'Under Construction (Possession: 2027)', '< 1 Year', or 'Ready to Move'**

Column 18: Age Evidence	**Data Source:** Contextual web snippet quoting property age, year built, or completion year
Underwriting Logic & Rules:	Captures exact snippet proof for property age and delivery year from real estate portals.
Mathematical Formula / Rule:	Contextual regex window
Display Format / Output:	**Verbatim snippet: '...Completed in 2023 (Age: 3 yrs)...' or '—'**

Column 19: Status & Age (Composite)	**Data Source:** Composite status and chronological age synthesis
Underwriting Logic & Rules:	Combines physical status with age duration into a single institutional underwriting tag.
Mathematical Formula / Rule:	f'{status_str} ({age_str})'
Display Format / Output:	**'Ready to Move (3 Years)', 'Under Construction (Possession: 2027)', or 'Ready to Move'**

Column 20: Scale & Typology	**Data Source:** Project land area (acreage), tower count, and unit count declarations
Underwriting Logic & Rules:	Classifies development into Township (>=20 acres or >=1000 units), Gated Community (3-20 acres or 4+ towers), Co-op CHS (40-200 units), or Standalone Building.
Mathematical Formula / Rule:	Hierarchical boundary check on acreage, tower, and unit counts
Display Format / Output:	**'Gated Community (6.2 Acres)', 'Township (150 Acres)', 'Standalone Building', or '—'**

Column 21: Scale Evidence	**Data Source:** Source portal snippet referencing land acreage, towers, or total units
Underwriting Logic & Rules:	Captures exact text declaration for community physical scale.
Mathematical Formula / Rule:	Contextual regex window
Display Format / Output:	**Verbatim snippet: '...Project spans 6.25 acres and comprises 8 towers...' or '—'**

Column 22: Evaluation Basis	**Data Source:** Elastic Valuation Engine mode selector
Underwriting Logic & Rules:	Indicates whether comparison was based on Full Valuation (Price + Physical specs), Physical Equivalence (Rate Unlisted), or Unverified.
Mathematical Formula / Rule:	If rate listed ➔ Full Valuation; Else if physical specs captured ➔ Physical Equivalence; Else ➔ Unverified
Display Format / Output:	**'Full Valuation Match (Price + Physical)', 'Physical / Structural Equivalence', or 'Unverified'**

Column 23: Valuation Match	**Data Source:** Continuous composite similarity score from Elastic Valuation Engine
Underwriting Logic & Rules:	Combines Gaussian price proximity, exponential distance decay, area tolerance, unit match, developer tier, age, and scale. Missing dimensions score 0.0.
Mathematical Formula / Rule:	$S_{total} = \Sigma (w_i \cdot S_i)$. Normal mode: 25% price + 15% dist + 15% area + 15% unit + 10% dev + 10% age + 10% scale.
Display Format / Output:	**Percentage string (e.g., '83.2%', '68.5%', '18.7%')**

Column 24: Status	**Data Source:** Strict Institutional Qualification Rules
Underwriting Logic & Rules:	Governs whether candidate is admitted into the valuation basket. Requires verified in-range rate OR verified in-range carpet area, and matching unit configuration.
Mathematical Formula / Rule:	If verified rate in 80%–120% and score >= 45% ➔ Qualified; If rate unlisted but area in ±15% and score >= 40% ➔ Qualified; If unit mismatch ➔ ■■ Unit Typology Mismatch.
Display Format / Output:	**'■ Qualified Match', '■■ Price Out of Range', '■■ Area Out of Range', '■■ Unit Typology Mismatch', or '■ Insufficient Data'**

Column 25: Evidence Citation	**Data Source:** Direct HTTP URL of the verified listing on MagicBricks, 99acres, Housing.com, or official developer portal
Underwriting Logic & Rules:	Provides direct clickable link to source evidence for human underwriter auditability.
Mathematical Formula / Rule:	Target URL string
Display Format / Output:	**Hyperlink: 'View Evidence ■'**

Column 26: Google Maps	**Data Source:** Google Place ID link generator
Underwriting Logic & Rules:	Deep-link to inspect location, satellite view, street view, and neighbourhood surroundings.
Mathematical Formula / Rule:	https://www.google.com/maps/place/?q=place_id:{place_id}
Display Format / Output:	**Hyperlink: 'Open Map ■'**

Column 27: Rating (Exported CSV)	**Data Source:** Google Places API (rating attribute)
Underwriting Logic & Rules:	The aggregate public star rating (scale 1.0 to 5.0) submitted by verified Google Maps contributors for the society.
Mathematical Formula / Rule:	Direct API float
Display Format / Output:	**Float (e.g., '4.2') or '0.0' if unrated**

Column 28: Reviews (Exported CSV)	**Data Source:** Google Places API (user_ratings_total attribute)
Underwriting Logic & Rules:	Total count of public reviews submitted on Google Maps. Used in underwriter audit trails to gauge society maturity and occupancy.
Mathematical Formula / Rule:	Direct API integer count
Display Format / Output:	**Integer count (e.g., '148') or '0'**

Column 29: Address (Exported CSV)	**Data Source:** Google Places API (vicinity or formatted_address)
Underwriting Logic & Rules:	Physical street address or localized municipal address string for the comparable candidate.
Mathematical Formula / Rule:	Cleaned address string
Display Format / Output:	**Address string (e.g., 'Sector 56A, Faridabad, Haryana 121004')**

4. The Elastic Valuation Engine & Mathematical Formulas

The engine calculates comparability using smooth continuous functions rather than arbitrary cutoffs. Missing attributes receive zero score (0.0) — never defaulted or artificial.

1. Price Proximity Score (S_price)

Evaluated using a Gaussian exponential decay model centered at Subject Rate R_src:
p_diff = |R_comp - R_src| / R_src
S_price = 100.0 · exp(-0.5 · (p_diff / 0.20)²)
• At ±0% diff ➔ 100.0 | At ±20% diff ➔ 60.65 | At ±40% diff ➔ 13.53

2. Spatial Proximity Score (S_dist)

Evaluated using exponential distance decay with characteristic scale length = 2.5 km:
S_dist = 100.0 · exp(-d / 2.5)
• At 0.0 km ➔ 100.0 | At 1.0 km ➔ 67.0 | At 2.5 km ➔ 36.8 | At 5.0 km ➔ 13.5

3. Carpet Area Score (S_area)

Evaluated using linear penalty over a 30% maximum divergence corridor:
a_diff = |A_comp - A_src| / max(A_src, 1.0)
S_area = max(0.0, 100.0 · (1.0 - a_diff / 0.30)) if both known;
• If candidate area unlisted ➔ 0.0 (Strict Zero Imputation)

4. Unit Typology Score (S_unit)

• Exact match with target unit (e.g., '2 BHK') ➔ 100.0
• Target unit missing, but other residential units present ➔ 30.0
• Undisclosed / unlisted unit typology ➔ 0.0

5. Developer Reputation Score (S_dev)

Based on Tier divergence ΔT = |Tier_comp - Tier_src|:
• Identical Tier (ΔT = 0) ➔ 100.0
• 1-Tier Gap (ΔT = 1, e.g. Tier 1 vs Tier 2) ➔ 65.0
• 2-Tier Gap (ΔT = 2, e.g. Tier 1 vs Tier 3) ➔ 30.0
• Undisclosed candidate developer ➔ 0.0

6. Status & Project Age Score (S_age)

Base score: Status_comp == Status_src ➔ 100.0; Status differs (RTM vs UC) ➔ 60.0
Age depreciation: **ΔAge = |Age_comp - Age_src|**
S_age = max(20.0, S_status_base - (ΔAge · 4.0))

7. Scale & Typology Score (S_scale)

• Identical scale classification ➔ 100.0
• Multi-acre interchange (Gated Community ↔ Township) ➔ 80.0
• Standalone Building ↔ Large Gated Complex ➔ 45.0
• Undisclosed candidate scale ➔ 0.0

4.2 Dynamic Elastic Weight Redistribution

When a candidate's rate is unlisted on public portals, the engine dynamically suppresses the price weight and elastically redistributes 100% of the scoring mass across physical equivalence dimensions:

Evaluation Dimension	Full Valuation Mode (Price Listed)	Physical Mode (Rate Unlisted)
Price Proximity (S_price)	25%	0% (Suppressed)
Spatial Proximity (S_dist)	15%	20%
Carpet Area Proximity (S_area)	15%	20%
Unit Typology Match (S_unit)	15%	20%
Developer Tier Match (S_dev)	10%	15%
Status & Age Match (S_age)	10%	15%
Scale & Typology Match (S_scale)	10%	10%
Total Weight Mass	**100%**	**100%**

5. Institutional Qualification & Summary KPI Metrics

5.1 Institutional Qualification Thresholds (Eliminating Unverified Noise)

Underwriting integrity requires that no project enters the qualified valuation basket without genuine evidence:
• ■ **Qualified Match:** (Verified rate in [0.80·R_src, 1.20·R_src] AND S_total >= 45.0%) OR (Rate unlisted, verified area in ±15%, AND S_total >= 40.0%).
• ■■ **Price Out of Range:** Verified rate exists but exceeds or falls below the 80%–120% boundary.
• ■■ **Area Out of Range:** Verified carpet area exists but deviates beyond ±15% from subject baseline.
• ■■ **Incomplete Specs (No Rate/Area):** Both rate and area unlisted; only minor metadata captured.
• ■ **Insufficient Data:** Zero verified attributes captured from public portals (raw proximity score 15%–20%).

5.2 Summary KPI Metrics & Outlier Statistics

Total Comparables Found	N_total = Count of all spatial entities within radius passing non-residential exclusion.
Qualified (All 4 Criteria)	N_qual = Count of projects meeting institutional qualification thresholds above.
Allowable Price Range	Corridor = [0.80 · R_src, 1.20 · R_src] anchoring the appraisal boundary.
Locality Cluster Median	Median = med({ R_comp | R_comp > 0 }), resilient to speculative portal spikes.
Interquartile Range (IQR)	IQR = Q3 - Q1 (75th percentile minus 25th percentile rate), measuring market dispersion.